Lecture plan · for the lecturer

Evaluating agents

AI agents · University · Lecture 3

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Define pass@k.
- Quantify uncertainty.
- Grade evaluation rigour.

Activity: Audit an evaluation

1. Warm-up (5 min): بأيّهما تثق؟
2. Compute (15 min): Pairs compute pass@1 and pass@3 from a sample of runs.
3. Interval (15 min): Add 95% intervals and decide which differences are real.
4. Audit (20 min): List what each team's report omits and rewrite one honestly.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- How wide was the interval?
- Which omission mattered most?
- How will you weight rigour?

Exit ticket

- Write the one sentence every agent result should include.

Try it in your next lecture
Show two teams' results, one inflated and one honest, without saying which; pairs list what each did and did not report, then vote on which they trust.

Seminar sheet · for students

Evaluating agents

AI agents · University · Lecture 3

Name _____ Class _____ Date _____

1 **Which result is most trustworthy for a student agent project?**

◯ pass@10 on development tasks ◯ Best of one run ◯ pass@1 over 5 runs on held-out tasks

2 **Compute the metrics.**

With p = 0.5, pass@3 = _____ . On 100 tasks at $\hat{p}$ = 0.5, the standard error is _____ .

Word bank: 0.875 · 0.75 · 0.5 · 0.05

3 **Which make an agent evaluation more trustworthy? Choose all that apply.**

☐ Several runs with an interval ☐ Tasks held out from development

☐ Reporting only the best run ☐ Reading transcripts of successes

4 **If agent A scores 62% and agent B 58% on 50 tasks in one run, A is clearly better.**

◯ True ◯ False

5 **Match each threat to evaluation validity with its remedy.**

Tuning on the test tasks	• • • •	Fresh, unpublished tasks
Run-to-run randomness	• • • •	Protected checkers and transcript review
Agent edits the tests	• • • •	Several runs and intervals
Benchmark in training data	• • • •	A held-out set

✎ **How do your students currently report results from agent or model projects, and what would you require next term?**